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United States Patent	12409246
Kind Code	B2
Date of Patent	September 09, 2025
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Aroma diffuser

Abstract

An aroma diffuser comprises a compression plug connected to an air pump, an oil tube, an atomizing chamber, a spray outlet base, and a rotary spray outlet cover. The atomizing chamber comprises a chamber, a spray outlet sleeve, and a spray outlet, a distance between the air outlet and a chamber wall of the atomizing chamber is 11-13 times an aperture diameter of the air outlet, the spray outlet of the atomizing chamber extends downward to form a spray outlet channel, the spray outlet sleeve is sleeved outside of the spray outlet channel, atomized essential oil enters into the gaps between the spray outlet sleeve and the spray outlet channel from the groove notch and is then sprayed out, a bottom of the spray outlet sleeve is obliquely disposed and comprises a first reflux hole, and the rotary spray outlet cover is rotatably connected to the spray outlet base.

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Appl. No.: 18/079145

Filed: December 12, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240115753 A1	Apr. 11, 2024

Foreign Application Priority Data

CN	202211230744.3	Oct. 08, 2022
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Publication Classification

Int. Cl.: A61L9/14 (20060101)

U.S. Cl.:

CPC A61L9/145 (20130101); A61L2209/134 (20130101)

Field of Classification Search

CPC: A61L (9/14)

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U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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2016/0000959	12/2015	Sevy	422/4	A61L 9/14

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
112656987	12/2020	CN	N/A

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Background/Summary

RELATED APPLICATIONS

(1) This application claims priority to Chinese patent application number 202211230744.3, filed on Oct. 8, 2022. Chinese patent application number 202211230744.3 is incorporated herein by reference.

FIELD OF THE DISCLOSURE

(2) The present disclosure relates to an aroma diffuser.

BACKGROUND OF THE DISCLOSURE

(3) At present, atomization of most aroma diffusers on the market is performed by sonicating with a transducer. Operating time of essential oil is short, and manufacturing cost is high. Some mechanical atomization structures lack control over an amount of the essential oil that is used, resulting in fast use of the essential oil. Chinese patent application with the publication number of CN112656987A provides an aroma diffuser. The aroma diffuser works according to Bernoulli's principle. Compressed air is accelerated by a compression plug to generate negative pressure, the essential oil in an essential oil bottle is sucked up through an oil tube and blown out to form a spray, and atomized essential oil is sprayed out of a spray outlet to achieve mechanical atomization effects. A blocking structure is disposed at an opening of an air outlet of the aroma diffuser, noise is large, and atomization is insufficient.

BRIEF SUMMARY OF THE DISCLOSURE

(4) The present disclosure provides an aroma diffuser to solve the deficiencies in the background.

(5) In order to solve the technical problem, a technical solution of the present disclosure is as follows.

(6) An aroma diffuser, which comprises a compression plug connected to an air pump, an oil tube, an atomizing chamber, a spray outlet base, and a rotary spray outlet cover. The oil tube and the atomizing chamber are connected to a bottle for receiving essential oil, air ejected from the compression plug is mixed with a liquid of the essential oil derived from the oil tube and then enters into the atomizing chamber through an air outlet, the atomizing chamber comprises a chamber, a spray outlet sleeve disposed in the chamber, and a spray outlet disposed on the chamber, a distance between the air outlet and a chamber wall of the atomizing chamber is 11-13 times an aperture diameter of the air outlet, the spray outlet of the atomizing chamber extends downward to form a spray outlet channel, the spray outlet sleeve is sleeved outside of the spray outlet channel, gaps are respectively disposed between an outer peripheral wall or a bottom of the spray outlet channel and the spray outlet sleeve, an upper part of the spray outlet sleeve comprises a groove notch, atomized essential oil enters into the gaps between the spray outlet sleeve and the spray outlet channel from the groove notch and is then sprayed out from the spray outlet channel and the spray outlet, a bottom of the spray outlet sleeve is obliquely disposed and comprises a first reflux hole at a lowest position, the spray outlet base is disposed on the spray outlet, the rotary spray outlet cover is rotatably connected to the spray outlet base, an atomizing transmitting chamber is formed between the spray outlet base and the rotary spray outlet cover, the spray outlet is in communication with the atomizing transmitting chamber, a bottom of the spray outlet base comprises a second reflux hole in communication with the atomizing chamber, and the rotary spray outlet cover comprises a spray outlet hole.

(7) In a preferred embodiment, the spray outlet sleeve comprises a spray baffle, and the spray baffle is staggered with the groove notch on a projection in a direction from a top of the spray outlet sleeve to the bottom of the spray outlet sleeve.

(8) In a preferred embodiment, the bottom of the spray outlet base comprises a reflux tube, and the reflux tube is in communication with the second reflux hole and the atomizing chamber.

(9) In a preferred embodiment, a spray transmission plate is disposed in the rotary spray outlet cover below the spray outlet hole, and the spray transmission plate is obliquely disposed relative to the spray outlet hole.

(10) In a preferred embodiment, an upper part of the spray outlet base is an annular wall, and the rotary spray outlet cover comprises an annular groove matched with the annular wall of the spray outlet base.

(11) In a preferred embodiment, an anti-detachment structure is disposed between the rotary spray outlet cover and the spray outlet base.

(12) In a preferred embodiment, the rotary spray outlet cover comprises an elastic snap joint configured to be clamped at the bottom of the spray outlet base.

(13) In a preferred embodiment, the chamber of the atomizing chamber comprises a pillar chamber part at a lower part and a tapered chamber part at an upper part that are connected to each other, and the spray outlet is disposed on a tapered top end of the tapered chamber part.

(14) In a preferred embodiment, a sealing structure is disposed between the pillar chamber part and the tapered chamber part.

(15) Compared with the existing techniques, the technical solution has the following advantages.

(16) The present disclosure provides gentle spray without large particles and reduced noise by maintaining the distance between the air outlet and the chamber wall of the atomizing chamber and the aperture diameter of the air outlet at a reasonable ratio, forming an atomizing channel between the spray outlet sleeve and the spray outlet channel, and forming the atomizing transmitting chamber between the spray outlet base and the rotary spray outlet cover. Noise is reduced to less than 43 dB. When the spray is produced, multi-layer reflux is achieved, the operating time of the essential oil is elongated, and rotary spray at 360° can be achieved.

- (17) One or more spray baffles are disposed on the spray outlet sleeve, the essential oil is transmitted, and ascending air current is blocked.
- (18) The anti-detachment structure is disposed between the rotary spray outlet cover and the spray outlet base, and the structure is reasonable.
- (19) The sealing structure is disposed between the polar chamber part and the tapered chamber part to further achieve hermetical sealing performance of the chamber.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a first cross-sectional view of an aroma diffuser of Embodiment 1 of the present disclosure.
- (2) FIG. 2 illustrates a second cross-sectional view of the aroma diffuser of Embodiment 1 of the present disclosure.
- (3) FIG. 3 illustrates an exploded perspective view of the aroma diffuser of Embodiment 1 of the present disclosure.
- (4) FIG. 4 illustrates an exploded view of a side surface of the aroma diffuser of Embodiment 1 of the present disclosure.
- (5) FIG. 5 illustrates a structural view of a spray outlet sleeve tube of the aroma diffuser of Embodiment 1 of the present disclosure.
- (6) FIG. 6 illustrates a top view of the spray outlet sleeve tube of the aroma diffuser of Embodiment 1 of the present disclosure.
- (7) FIG. 7 illustrates a cross-sectional view of an aroma diffuser of Embodiment 2 of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(8) The present disclosure will be further described below in combination with the accompanying drawings and embodiments. However, the aroma diffuser of the present disclosure is not limited to the embodiments.

Embodiment 1

- (9) Referring to FIGS. 1 to 6, an aroma diffuser of the present disclosure comprises a compression plug 1 connected to an air pump, an oil tube 21, and an atomizing chamber 3 connected to a bottle 2 for receiving essential oil. Air ejected from the compression plug 1 is mixed with a liquid of the essential oil derived from the oil tube 21 and then enters the atomizing chamber 3 through an air outlet 4.
- (10) The atomizing chamber 3 comprises a chamber, a spray outlet sleeve tube 33 disposed in the chamber, and a spray outlet 34 disposed on the chamber. In this embodiment, the chamber of the atomizing chamber 3 comprises a pillar chamber part 31 at a lower side and a tapered chamber part 32 (e.g., a conical chamber part) at an upper side that are connected each other. A sealing structure, such as an O-ring 35, is disposed between the pillar chamber part 31 and the tapered chamber part 32.
- (11) In this embodiment, a distance between the air outlet 4 and a chamber wall of the chamber (the pillar chamber part 31) of the atomizing chamber is 11-13 times an aperture diameter of the air outlet 4. Therefore, the aroma diffuser in this embodiment can achieve good atomization effects, sufficient atomization, and small atomization particle size.
- (12) The spray outlet 34 is disposed on a tapered top end of the tapered chamber part 32. The spray outlet 34 extends downward to form a spray outlet channel 36, and the spray outlet sleeve tube 33 is sleeved outside of the spray outlet channel 36. Referring to FIG. 1, gaps are respectively disposed between an outer peripheral wall or a bottom of the spray outlet channel 36 and the spray outlet sleeve tube 33. Referring in combination with FIGS. 5 and 6, an upper part of the spray

outlet sleeve tube **33** comprises two groove notches **331**, and middle and lower parts of the spray outlet sleeve tube **33** comprise two spray baffles **332**. The two spray baffles **332** and the two groove notches **331** are staggered on a projection surface, and the two spray baffles **332** are used to transmit the essential oil and block ascending air current. Atomized essential oil enters into the gaps between the spray outlet sleeve tube **33** and the spray outlet channel **36** from the two groove notches **331** and is then sprayed out from the spray outlet channel **36** and the spray outlet **34**. The bottom of the spray outlet sleeve tube **33** is inclined and comprises a first reflux hole **333** at a lowest position. The aroma diffuser further comprises a spray outlet base **5** and a rotary spray outlet cover **6**. The spray outlet base **5** is disposed on the spray outlet **34**, and the rotary spray outlet cover **6** is rotatably connected to the spray outlet base **5**. The spray outlet base **5** and the rotary spray outlet cover **6** form an atomizing transmitting chamber **60**, and the spray outlet **34** is in communication with the atomizing transmitting chamber **60**. A bottom of the spray outlet base **5** comprises a second reflux hole **51** in communication with the atomizing chamber **3**. The rotary spray outlet cover **6** comprises a spray outlet hole **61**, so the rotary spray outlet cover **6** can be rotated to achieve rotary spray within 360° by operation.

(13) Referring to FIGS. **2** and **3**, the bottom of the spray outlet base **5** comprises a reflux tube **52**, and the reflux tube **52** is in communication with the second reflux hole **51** and the atomizing chamber **3**. Furthermore, a middle of the bottom of the spray outlet base **5** comprises a protrusion that matches a top of the tapered chamber part **32**. A bottom wall inside the spray outlet base **5** forms an annular groove, and the second reflux hole **51** is disposed on the annular groove to facilitate recovery and transmission of the essential oil.

(14) A spray transmission plate **7** is disposed in the rotary spray outlet cover **6** below the spray outlet hole **61**, and the spray transmission plate **7** is obliquely disposed relative to the spray outlet hole **61** to achieve oblique spray effects.

(15) Specifically, referring to FIGS. **1** and **3**, an upper part of the spray outlet base **5** is an annular wall, and the rotary spray outlet cover **6** comprises an annular groove matched with the annular wall of the spray outlet base **5**. At the same time, an anti-detachment structure is disposed between the rotary spray outlet cover **6** and the spray outlet base **5**. Specifically, the rotary spray outlet cover **6** comprises an elastic snap joint **62** that can be clamped at the bottom of the spray outlet base **5**.

Embodiment 2

(16) Referring to FIG. **7**, Embodiment 2 differs from Embodiment 1 in that Embodiment 2 is not provided with the spray transmission plate **7**, and the spray is directly ejected from the spray outlet hole **61** in the atomizing transmitting chamber **60**.

(17) The aforementioned embodiments are merely some embodiments of the aroma diffuser of the present disclosure, and the scope of the disclosure is not limited thereto. Thus, it is intended that the present disclosure cover any modifications and variations of the presently presented embodiments provided they are made without departing from the appended claims and the specification of the present disclosure.

Claims

1. An aroma diffuser, comprising: a compression plug connected to an air pump, an oil tube, an atomizing chamber, a spray outlet base, and a rotary spray outlet cover, wherein: the oil tube and the atomizing chamber are connected to a bottle for receiving essential oil, air ejected from the compression plug is mixed with a liquid of the essential oil derived from the oil tube and then enters into the atomizing chamber through an air outlet, the atomizing chamber comprises a chamber, a spray outlet sleeve disposed in the chamber, and a spray outlet disposed on the chamber, a distance between the air outlet and a chamber wall of the atomizing chamber is 11-13 times an aperture diameter of the air outlet, the spray outlet of the atomizing chamber extends downward to form a spray outlet channel, the spray outlet sleeve is sleeved outside of the spray outlet channel,

gaps are respectively disposed between an outer peripheral wall or a bottom of the spray outlet channel and the spray outlet sleeve, an upper part of the spray outlet sleeve comprises a groove notch, atomized essential oil enters into the gaps between the spray outlet sleeve and the spray outlet channel from the groove notch and is then sprayed out from the spray outlet channel and the spray outlet, a bottom of the spray outlet sleeve is obliquely disposed and comprises a first reflux hole at a lowest position, the spray outlet base is disposed on the spray outlet, the rotary spray outlet cover is rotatably connected to the spray outlet base, an atomizing transmitting chamber is formed between the spray outlet base and the rotary spray outlet cover, the spray outlet is in communication with the atomizing transmitting chamber, a bottom of the spray outlet base comprises a second reflux hole in communication with the atomizing chamber, and the rotary spray outlet cover comprises a spray outlet hole.

2. The aroma diffuser according to claim 1, wherein: the spray outlet sleeve comprises a spray baffle, and the spray baffle is staggered with the groove notch on a projection in a direction from a top of the spray outlet sleeve to the bottom of the spray outlet sleeve.
 3. The aroma diffuser according to claim 1, wherein: the bottom of the spray outlet base comprises a reflux tube, and the reflux tube is in communication with the second reflux hole and the atomizing chamber.
 4. The aroma diffuser according to claim 1, wherein: a spray transmission plate is disposed in the rotary spray outlet cover below the spray outlet hole, and the spray transmission plate is obliquely disposed relative to the spray outlet hole.
 5. The aroma diffuser according to claim 1, wherein: an upper part of the spray outlet base is an annular wall, and the rotary spray outlet cover comprises an annular groove matched with the annular wall of the spray outlet base.
 6. The aroma diffuser according to claim 1, wherein an anti-detachment structure is disposed between the rotary spray outlet cover and the spray outlet base.
 7. The aroma diffuser according to claim 1, wherein the rotary spray outlet cover comprises an elastic snap joint configured to be clamped at the bottom of the spray outlet base.
 8. The aroma diffuser according to claim 1, wherein: the chamber of the atomizing chamber comprises a pillar chamber part at a lower part and a tapered chamber part at an upper part that are connected each other, and the spray outlet is disposed on a tapered top end of the tapered chamber part.
 9. The aroma diffuser according to claim 8, wherein a sealing structure is disposed between the pillar chamber part and the tapered chamber part.
 10. The aroma diffuser according to claim 5, wherein an anti-detachment structure is disposed between the rotary spray outlet cover and the spray outlet base.
 11. The aroma diffuser according to claim 5, wherein the rotary spray outlet cover comprises an elastic snap joint configured to be clamped at the bottom of the spray outlet base.
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